

25.10.2023

Matematik strukturalar. Gruppya,
xalqa, maydon va uning xossalari.

Kriptografik algoritmlar va proto-
kollar ma'lumotni son yoki chekli
fazo elementlari sifatida qayta
ishlaydi. Maxsus holda shifrlash
va ochiq matnga ega bo'lish jarro
(deshifrlash) jarayonida foydalani-
ladigan amallar berilgan ma'lumot-
ni bir tordan boshqa turga otkara-
di hamda chekli fazoda yopiqlik
(binar amal) xossasiga ega bo'lishli
loxim. Biroq sonlar ustidagi oddiy
arifmetik amallar (+, -, *, /) ushbu
xossani bojarmaydi. Shuning uchun
kriptografik algoritmlarda chekli fa-
zodagi ma'lumotlar amalda sonlar
ustidagi oddiy arifmetik amallarga
keltirilmaydi. Qoidaga asosan

ularada algebraik struktura aniqlangan
va yopiqlik xossasini bajaruvchi fanoalarda
amalga oshadi.

Ta'rif: Ixtiyoriy A to'plamning tayin
bir tartibda olingan $\forall x, y$ - elementiga
shu to'plamning aniq bir elementi
mos qoyilsa, A to'plamda binar algeb-
raik amal aniqlangan deyiladi.

Binar amal:

$\cdot, 0, \tau, \oplus, \dots$ ($*$ - binar to'plam)
kabi belgilanadi.

$a \otimes b = ? \rightarrow a$ ni b ga R parametrlili qo'shish.

Ta'rif: Aytaylik A to'plamda biror $*$
binar amal aniqlangan bolsin. Agar
 $\forall x, y, z \in A$ uchun $(x * y) * z = x * (y * z)$ bolsa,
 $*$ binar amal **assosiativ** d-di.

Ta'rif: Agar $\forall x, y \in A$ uchun $x * y = y * x$
bolsa, $*$ binar amal **kommutativ** deyiladi.

Ta'rif: Agar quyidagi shartlar bajarilsa:

1. G toplamda binar amal $*$ aniqlangan;

2. Binar amal $*$, assosiativ:

$$\forall x, y, z \in G \text{ uchun } (x * y) * z = x * (y * z)$$

3. G -da neytral element " e " mavjud:

$$\forall a \in G \text{ uchun } a * e = e * a = a \text{ shuni bo'lsa}$$

4. $\forall a \in G$ uchun G -da a -ga simmetrik

(teskari) element mavjud:

$$a * a^{-1} = a^{-1} * a = e$$

bu G toplam **grupp**a deyiladi.

Ta'rif: Agar grupp ta'rifida 1- va 2-shartlar bajarilsa G toplamga **yarim grupp**a deyiladi.

Ta'rif: Agar grupp ta'rifida, 1-, 2- va 3-shartlar bajarilsa, u holda bu toplamga **monoid** deyiladi.

Grupp terminini fanga 1-bolib, fransuz matematigi Galua 1832-yilda kiritgan.

Misol.

Bütün sonlar toplami qo'shish amaliga nisbatan gruppni tashkil etadi.

Tarif: K toplamda qo'shish va (~~ayirish~~) ko'paytirish amallari aniqlangan bo'lib, quyidagi shartlar bajarilgan bo'lsa:

1) K -toplam qo'shish amaliga nisbatan Abel gruppasi tashkil qilsa,

2) Ko'paytirish amali assosiativ.

3) Ko'paytirish amali qo'shish amaliga nisbatan distributiv, ya'ni

$$\forall a, b, c \text{ uchun } a \cdot (b + c) = ab + ac$$

U holda $(K, +, \cdot)$ struktura

halqa deb ataladi.

Misol 2.

Bütün sonlar toplamida $+$ va $\times$ xalqa tashkil etadi.

Misol 3

$(n \times n)$ A-matritsa toplam berilgan.

"." - ko'paytirish amali. (binar)
qachon grupp tashkil etadi.

Vazifa.

Misol 1. Yechish: $a, b, c \in \mathbb{Z}$ $(+)$ $\rightarrow$ binar amal

Grupp:

- 1) Binar $(+)$ amal aniqlangan.
- 2) $(a+b)+c = a+(b+c) \rightarrow$ assosiativ.
- 3) $e=0, e \in \mathbb{Z}$
 $a+0 = 0+a = a$
- 4) $a+(-a) = (-a)+a = 0$

Misol 2. Yechish

$(+)$ amali, $(*)$ amali

- 1) Bütün sonlar toplamida aniqlangan $(+, *)$
- 2) Assosiativ.
 $(a+b)+c = a+(b+c), \quad (a*b)*c = a*(b*c)$

3) $(+)$ uchun $e=0 \quad a+0 = 0+a = a$

(*) uchun $e=1$ $a \times 1 = 1 \times a = a$.
 Demak (*) va (+) amallari butun sonlar to'plamida
 xalqa tashkil etadi.

Misol 3.

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} \quad (*) - \text{binar amal.}$$

1) (*) amali aniqlangan.

2) Assosiativlik $A \star (B \cdot C) = (A \cdot B) \cdot C$

$$A \cdot B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \cdot \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 5+14 & 6+16 \\ 15+28 & 18+32 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

$$B \cdot C = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \cdot \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 5+12 & 15+24 \\ 7+16 & 21+32 \end{pmatrix} = \begin{pmatrix} 17 & 39 \\ 23 & 53 \end{pmatrix}$$

$$A \cdot (B \cdot C) = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \cdot \begin{pmatrix} 17 & 39 \\ 23 & 53 \end{pmatrix} = \begin{pmatrix} 17+46 & 39+106 \\ 51+92 & 117+212 \end{pmatrix} = \begin{pmatrix} 63 & 145 \\ 143 & 329 \end{pmatrix}$$

$$(A \cdot B) \cdot C = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix} \cdot \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 19+44 & 57+88 \\ 43+100 & 129+200 \end{pmatrix} = \begin{pmatrix} 63 & 145 \\ 143 & 329 \end{pmatrix}$$

3) $e = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $A \cdot e = e \cdot A = A$

4) $A \cdot A^{-1} = A^{-1} \cdot A = e$ (Agar $\det A \neq 0$ bo'lsa)

Misol 2. (+) va (*) xalqa

1) $\mathbb{Z}$ to'plam (+) amaliga nisbatan Abel gruppasi
 tashkil etadi.

2) (*) - assosiativ $(a \star b) \star c = a \star (b \star c)$

3) distributiv $a(b+c) = a \cdot b + a \cdot c$

4) $(\mathbb{Z}, +, \times)$ struktura xalqa tashkil
 etadi.

KueB
 2. XI. 23r.

30.10.2023.

AES algoritmining matematik asoslari

Xalqaning quyidagi xossalari mavjud.

1. Har bir xalqada uning noli deb ataluvchi, yagona θ -element mavjud.

2. Har bir xalqa ixtiyoriy elementi uchun unga qarama-qarshi bolgan yagona element mavjud, ya'ni $\forall a \in K$ uchun shunday " $-a$ " element topiladiki $a + (-a) = \theta$

3. Xalqada (K) $a + x = b$ tenglama yechimga ega va u yagonadir. Bu yechim

$$x = -a + b$$

ga teng

Ta'rif: Agar K -xalqaning ixtiyoriy a va b elementlari uchun $ab = ba$ tenglik bajarilsa, u holda K -xalqa **kommutativ xalqa** deyiladi.

Ta'rif: P -kommutativ xalqa bolsin. Agar $\forall a, b \in P$ va $a \neq 0$ uchun $ax = b$ tenglikni qanoatlantiruvchi faqat bitta element $x \in P$ mavjud bolsa, u holda P -to'plam maydon deyiladi.

Maydon xossalari.

1. Har bir P maydonda birlik element mavjud va u yagona.
2. $x \neq 0 \in P$ bolsa, P -da x -ga teskari yagona x^{-1} element mavjud.
3. Ixtiyoriy P maydon nolining boluvchilariga (nisbatan) ega emas. Ya'ni $\forall x, y \in P$ va $xy = 0$ bolsa, u holda $x = 0$ yoki $y = 0$.

Misol. $f(x) = \frac{a_1x+b_1}{a_2x+d_2}$ - kasr chiziqli funksiyalar toplami.

M_f -b-u belgilangan binar amal, quyidagi: $f_1 \circ f_2 = f_1(f_2(x))$ bolsa, u holda toplanm gruppami?

Misol 2. Kompleks sonlar toplami odatdagi qo'shish va ko'paytirish amaliga nisbatan maydon tashkil etishi ko'rsatilsin.

— Kvadrat matritsalar kommutativlik xossasiga ega emas, lekin assosiativlik xossasiga ega.

Kom: $a * b \neq b * a$

Assos: $a * (b * c) = (a * b) * c$.

Ma'ru: Keltirilmaydigan ko'phadlar maydoni.

Sub Bytes() $\Leftrightarrow \begin{cases} S_{ij} \cdot S_{ij}^{-1} \equiv 1 \pmod{\varphi(x)}, \varphi(x) \in GF(2^8) \\ S_{ij}^{-1} = b_i, c \oplus b_1 = \text{bayt} \end{cases}$

keltirilmaydigan 8-darajali ko'phad.

Tarif: $g(x) \in F(x)$ - kophad $f(x) \in F(x)$ kophad-
ni boladi deyiladi, agar $\exists h(x) \in F(x)$ - kophad
ma'jud bo'lib, $f(x) = g(x) \cdot h(x)$ bo'linli bo'lsa.
Bu yerda $F(x)$ - qandaydir F -maydonda
kophadlar to'plami deb qaraladi.

Misol 1. $f(x) = 2x^5 + x^4 + 4x + 3$ $g(x) = 3x^2 + 1$ ($\in F_5[x]$)
 $q(x) = ?$ $r(x) = ?$ ($\in F_5[x] = ?$)

$$\begin{array}{r}
 2x^5 + x^4 + 4x + 3 \quad | \quad 3x^2 + 1 \\
 - 2x^5 + \quad 4x^3 \quad \quad \quad 4x^3 + 2x^2 + 2x + 1 \\
 \hline
 x^4 - 4x^3 + 4x + 3 \\
 = x^4 + x^3 + 4x + 3 \\
 - x^4 + \quad 2x^2 \quad \quad \quad \\
 \hline
 x^3 - 2x^2 + 4x + 3 \\
 = x^3 + 3x^2 + 4x + 3 \\
 - x^3 + \quad 2x \quad \quad \quad \\
 \hline
 - 3x^2 + 2x + 3 \\
 \quad 3x^2 + \quad 1 \\
 \hline
 \quad \quad 2x + 2
 \end{array}$$

$$r(x) = 2x + 2$$

$$q(x) = 4x^3 + 2x^2 + 2x + 1$$

Varifa

Misol 3

Korsatilpen maydonda Yevklod usuli
bilan $\text{EKUB}(f, g)$ topilsin.

A) $f = x^7 + 1$, $g(x) = x^5 + x^3 + x + 1$, $F = F_2$

B) $f = x^5 + x + 1$, $g(x) = x^6 + x^5 + x^4 + 1$, $F = F_2$

C) $f = x^8 + 2x^5 + x^3 + x^2 + 1$, $g(x) = 2x^6 + x^5 + 2x^3 + 2x^2 + 2$, $F = F_3$

Vazifa Yechish.

Misol 2. kompleks sonlar $(+)$, $(*)$ - binar amallar
Maydon tashkil etishi.

1) kompleks sonlar maydonida $i\sqrt{-1}$
birik element h-di.

2) A) $x = 2i + 3 \neq 0$

$$x = a + bi$$

$$x^{-1} = \frac{1}{2i+3}$$

$$x^{-1} = \frac{1}{a+bi}$$

A) $\Rightarrow x^{-1} = -x$

3) kompleks sonlar maydonining
holi $b=0 \rightarrow x = a+bi = 0$. $\theta=0$
 $a=0 \Rightarrow$

$x \cdot y = 0$ bolishi uchun x yoki y
holga teng bolishi kerak.

Demak kompleks sonlar maydoni Maydon
tashkil etadi.

Misol 3.

A) $f(x) = x^7 + 1$, $g(x) = x^5 + x^3 + x + 1$ $F = F_2$

$$\begin{array}{r} x^7 + 1 \mid x^5 + x^3 + x + 1 \\ -x^5 + x^3 + x^2 + 1 \\ \hline -x^5 - x^3 - x^2 + 1 \end{array}$$

mod 2 $\rightarrow x^5 + x^3 + x^2 + 1$

$$\begin{array}{r} -x^5 + x^3 + x^2 + 1 \\ \hline x^2 - x \end{array}$$

mod 2 $\rightarrow (x^2 + x)$

$$\begin{array}{r} x^5 + x^3 + x + 1 \mid x^2 + x \\ -x^5 + x^4 \\ \hline -x^4 + x^3 + x + 1 \end{array}$$

mod 2 $\rightarrow x^4 + x^3 + x + 1$

$$\begin{array}{r} -x^4 + x^3 \\ \hline x^3 + x + 1 \end{array}$$

$(x+1) \rightarrow \text{EKUB } (f(x), g(x))$

$$\begin{array}{r} x^2 + x \mid x + 1 \\ -x^2 + x \\ \hline 0 \end{array}$$

$$b) f(x) = x^5 + x + 1, \quad g(x) = x + x^2 + x^3 + 1, \quad F = F_2$$

$$\begin{array}{r} x^6 + x^5 + x^4 + 1 \quad | \quad x^5 + x + 1 \\ - x^6 + x^2 + x \\ \hline x^5 - x^4 - x^2 - x + 1 \\ \rightarrow - x^5 + x^4 + x^2 + x + 1 \\ \hline x^4 + x^2 \end{array}$$

$$\begin{array}{r} x^5 + x + 1 \quad | \quad x^4 + x^2 \\ - x^5 + x^3 \\ \hline x^3 + x + 1 \end{array}$$

$$\begin{array}{r} x^4 + x^2 \quad | \quad x^3 + x + 1 \\ - x^4 + x^2 + x \\ \hline x \end{array}$$

$$\begin{array}{r} x^3 + x + 1 \quad | \quad x \\ - x^3 \\ \hline x + 1 \\ - x \\ \hline 1 \end{array}$$

$f(x)$ va $g(x)$ o'zaro tub.

$$c) f(x) = x^8 + 2x^5 + x^3 + x^2 + 1 \quad g(x) = 2x^6 + x^5 + 2x^3 + 2x^2 + 2 \quad F = F_3$$

$$\begin{array}{r} x^8 + 2x^5 + x^3 + x^2 + 1 \quad | \quad 2x^6 + x^5 + 2x^3 + 2x^2 + 2 \\ - 4x^8 + 2x^7 + 4x^5 + 4x^4 + 4x^3 \\ \hline 3x^7 - 2x^5 - 4x^4 + x^3 + 3x^2 + 1 \end{array}$$

$$\rightarrow \begin{array}{r} x^7 + x^5 + 2x^4 + x^3 + 1 \\ - 4x^7 + 2x^6 + 4x^4 + 4x^3 + 4x \\ \hline -3x^7 - 2x^6 + x^5 - 2x^4 - 3x^3 - 4x + 1 \end{array}$$

$$\rightarrow \begin{array}{r} x^6 + x^5 + x^4 + 2x + 1 \\ - 4x^6 + 2x^5 + 4x^3 + 4x^2 + 4 \\ \hline -3x^6 - x^5 + x^4 - 4x^3 - 4x^2 + 2x - 3 \end{array}$$

$$\rightarrow \begin{array}{r} 2x^5 + x^4 + 2x^3 + 2x^2 + 2x \\ - 8x^6 + x^5 + 2x^4 + 2x^3 + 2x^2 \\ \hline -2x^4 + 2 \end{array}$$

$$\rightarrow \begin{array}{r} 2x^5 + x^4 + 2x^3 + 2x^2 + 2x \\ - 8x^6 + x^5 + 2x^4 + 2x^3 + 2x^2 \\ \hline -2x^4 + 2 \end{array}$$

$$\rightarrow \begin{array}{r} 2x^5 + x^4 + 2x^3 + 2x^2 + 2x \\ - 8x^6 + x^5 + 2x^4 + 2x^3 + 2x^2 \\ \hline -2x^4 + 2 \end{array}$$

$$\rightarrow \begin{array}{r} 2x^5 + x^4 + 2x^3 + 2x^2 + 2x \\ - 8x^6 + x^5 + 2x^4 + 2x^3 + 2x^2 \\ \hline -2x^4 + 2 \end{array}$$

$$\rightarrow 2x^5 + x^4 + 2x^3 + 2x^2 + 2x$$

$$\begin{array}{r} 2x^6 + x^5 + 2x^3 + 2x^2 + 2 \quad | \quad 2x^5 + x^4 + 2x^3 + 2x^2 + 2x \\ - 8x^6 + x^5 + 2x^4 + 2x^3 + 2x^2 \\ \hline -2x^4 + 2 \end{array}$$

$$\text{mod } 3 \rightarrow x^4 + 2$$

Kurd

2. XI. 23u.

$$\begin{array}{r}
 2x^5 + x^4 + 2x^3 + 2x^2 + 2x \quad | \quad x^4 + 2 \\
 - 2x^5 + 4x \\
 \hline
 x^4 + 2x^3 + 2x^2 - 2x \\
 - x^4 + 2 \\
 \hline
 2x^3 + 2x^2 + x + 1
 \end{array}$$

$$\begin{array}{r}
 x^4 + 2 \quad | \quad 2x^3 + 2x^2 + x + 1 \\
 + 2x^4 + 4x^3 + 2x^2 + 4x \\
 \hline
 -3x^3 - 2x^2 - 2x + 1
 \end{array}$$

$$\text{mod } 3 \rightarrow 2x^2 + x$$

$$\begin{array}{r}
 2x^3 + 2x^2 + x + 1 \quad | \quad 2x^2 + x \\
 - 2x^3 + x^2 \\
 \hline
 x^2 + x + 1 \\
 - x^2 + x \\
 \hline
 1 \\
 - 3x^2 - x + 1
 \end{array}$$

$$\text{mod } 3 \rightarrow \boxed{2x + 1} \rightarrow \text{EKUB}(f(x), g(x))$$

$$\begin{array}{r}
 2x^2 + x \quad | \quad 2x + 1 \\
 - 2x^2 + x \\
 \hline
 0
 \end{array}$$

$$\begin{array}{r}
 \text{mod } 3 \quad 2x^3 + x^2 + x + 2 \\
 2x^3 + 2x^2 + x + 1 \\
 \hline
 -x^2 + 1
 \end{array}$$

$$\text{mod } 3 \rightarrow \boxed{2x^2 + 1}$$

$$\begin{array}{r}
 2x^3 + 2x^2 + x + 1 \quad | \quad 2x^2 + 1 \\
 - 2x^3 + x \\
 \hline
 x^2 + x + 1 \\
 - x^2 + 1 \\
 \hline
 x
 \end{array}$$

$$\text{EKUB}(f(x), g(x)) = 2x^2 + 1$$

Tarifi: $f(x) \in F[x]$ ko'phad keltiriladigan ko'phad deb atiladi. F -maydon ustida yoki $F[x]$ -xalqada, agar $f(x) = g \cdot h$; $g, h \in F(x)$ tenglik bajariladigan faqat $g = \text{const}$ yoki $h = \text{const}$ bo'lganda, ya'ni o'zgarmas ko'phadlar deb tushuniladi.

Misol. F_2 maydondagi 4-darajali barcha keltirilmaydigan ko'phadlarni toping.

$$f_1 = x^4 + x + 1 \quad f_2(x) = x^4 + x^3 + 1$$

$$f_3(x) = x^4 + x^3 + x^2 + x$$

shu 3 ta ko'phad keltirilmaydigan maydon.

- 1) x^4
- 2) $x^4 + x + 1$
- 3) $x^4 + x^2 + x + 1$
- 4) $x^4 + x^3 + x^2 + x + 1$
- 5) $x^4 + 1 \rightarrow (x+1)$ ga bo'linadi.

$$(a_0 + a_1x + a_2x^2 + x^3) \cdot (b_0 + x) \quad \text{— shu 2 ta ko'rimishdan biri bo'ladi}$$

$$(a_0 + a_1x + x^2)(b_0 + b_1x + x^2)$$

Teorema: Faraz qilaylik $f(x) \in F[x]$,

$F[x]/f(x)$ -faktor xalqa maydon bolishligi

uchun $f(x)$ - kophad F -maydonda keltirilgan
magan bolishligi zarur va yetarli.

Teorema. $f(x) \in F[x]$ kophad keltirilmay-
digan 2- va 3- darajali bolishligi uchun
 $F[x]$ xalqada, $f(x)$ - kophad F -maydonda
ildizga ega bolmasligi zarur va
yetarlidir.

Misol. Faraz qilaylik $f(x) = x^2 + x + 1 \in F[x]$.
U holda faktor halqa $F_2[x]/(f)$ - quy-
dagi elementdan $p^2 = 2^2$ yoki $0, 1, x, x+1$
iborat.

"+"	0	1	x	x+1
0	0	1	x	x+1
1	1	0	x+1	x
x	x	x+1	0	1
x+1	x+1	x	1	0

"·"	0	1	x	x+1
0	0	0	0	0
1	0	1	x	x+1
x	0	x	x+1	1
x+1	0	x+1	1	x

$$[x+1] \cdot [x] = x^2 + x$$

$$[x+1] \cdot [x+1] = x^2 + 2x + 1$$

$$\begin{array}{r|l} x^2 + x & x^2 + x + 1 \\ -x^2 + x + 1 & 1 \\ \hline -1 & \\ \text{mod } 2 & [1] \end{array}$$

$$\begin{array}{r|l} x^2 + 2x + 1 & x^2 + x + 1 \\ -x^2 + x + 1 & 1 \\ \hline [x] & \end{array}$$

amirast

Quyidagi $f_2(x) / (x^3 + x^2 + x)$ faktor xalga
 bo'linish va ko'paytirish jadvalлари.
 tekshirilsin va uning maydon tashkil
 qilishini tekshirilsin.

Misol 1.

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + x^4, \quad F = F_2$$

a_0	a_1	a_2	a_3
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

$$\begin{aligned}
 f_1(x) &= x^4 = x^2 \cdot x^2 = x \cdot x^3 \\
 f_2(x) &= x^4 + x^3 = x(x^3 + x^2) = x^2(x + x^2) = x^2 f_1(x) \\
 f_3(x) &= x^4 + x^2 = x(x^3 + x) = x^2(x^2 + 1) \\
 f_4(x) &= x^4 + x^3 + x^2 = x^2(x^2 + x + 1) \\
 f_5(x) &= x^4 + x = x(x^3 + 1) \\
 f_6(x) &= x^4 + x^3 + x = x \cdot (x^3 + x^2 + 1) \\
 f_7(x) &= x^4 + x^2 + x = x \cdot (x^3 + x + 1) \\
 f_8(x) &= x^4 + x^3 + x^2 + x = x \cdot (x^3 + x^2 + x + 1) \\
 f_9(x) &= x^4 + 1 \\
 f_{10}(x) &= x^4 + x^3 + 1 \\
 f_{11}(x) &= x^4 + x^2 + 1 \\
 f_{12}(x) &= x^4 + x^3 + x^2 + 1 \\
 f_{13}(x) &= x^4 + x + 1 \\
 f_{14}(x) &= x^4 + x^3 + x + 1 \\
 f_{15}(x) &= x^4 + x^2 + x + 1 \\
 f_{16}(x) &= x^4 + x^3 + x^2 + x + 1
 \end{aligned}$$

$$f_9(x) = x^4 + 1$$

$$\begin{array}{r}
 x^4 + 1 \mid x + 1 \\
 \underline{x^4 + x^3} \\
 -x^3 + 1 \\
 \text{mod } 2 \rightarrow \underline{x^3 + 1} \\
 \underline{x^3 + x^2} \\
 -x^2 + 1 \\
 \text{mod } 2 \rightarrow \underline{x^2 + 1}
 \end{array}$$

$$\begin{array}{r}
 x^4 + 1 \\
 \underline{x^3 + x} \\
 -x + 1 \\
 \text{mod } 2 \rightarrow \underline{x + 1} \\
 \underline{x + 1} \\
 0
 \end{array}$$

$$f_9(x) = x^4 + 1 = (x+1) \cdot (x^3 + x^2 + x + 1)$$

$$- f_{10}(x) = x^4 + x^3 + 1$$

$$\begin{array}{r|l} x^4 + x^3 + 1 & x+1 \\ \underline{x^4 + x^3} & \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} x^4 + x^3 + 1 & x^2 + x + 1 \\ \underline{x^4 + x^3 + x^2 + x + 1} & \\ \hline & 0 \end{array}$$

$$\begin{array}{r|l} x^4 + x^3 + 1 & x^2 + x + 1 \\ \underline{x^4 + x^3 + x^2 + x + 1} & \\ \hline & 0 \end{array}$$

$f_{10}(x) = x^4 + x^3 + 1 \rightarrow$ keltirilmaydigan köphad.

$$- f_{11}(x) = x^4 + x^2 + 1$$

$$\begin{array}{r|l} x^4 + x^2 + 1 & x+1 \\ \underline{x^4 + x^3} & \\ \hline & -x^3 + x^2 + 1 \end{array}$$

mod 2 $\rightarrow$

$$\begin{array}{r|l} x^3 + x^2 & \\ \underline{x^3 + x^2} & \\ \hline & 0 \end{array}$$

$$\begin{array}{r|l} x^4 + x^2 + 1 & x^2 + x + 1 \\ \underline{x^4 + x^3 + x^2} & \\ \hline & x^3 + 1 \end{array}$$

$$\begin{array}{r|l} x^3 + 1 & x^2 + x + 1 \\ \underline{x^3 + x^2 + x} & \\ \hline & -x^2 + x + 1 \end{array}$$

$$\begin{array}{r|l} -x^2 + x + 1 & \\ \underline{-x^2 + x + 1} & \\ \hline & 0 \end{array}$$

$$f_{11}(x) = x^4 + x^2 + 1 = (x^2 + x + 1) \cdot (x^2 + x + 1)$$

$$- f_{12}(x) = x^4 + x^3 + x^2 + 1$$

$$\begin{array}{r|l} x^4 + x^3 + x^2 + 1 & x+1 \\ \underline{x^4 + x^3} & \\ \hline & x^2 + 1 \end{array}$$

$$\begin{array}{r|l} x^2 + 1 & \\ \underline{x^2 + x} & \\ \hline & -x + 1 \end{array}$$

mod 2 $\rightarrow$

$$\begin{array}{r|l} -x + 1 & \\ \underline{x + 1} & \\ \hline & 0 \end{array}$$

$$f_{12}(x) = x^4 + x^3 + x^2 + 1 = (x+1) \cdot (x^3 + x + 1)$$

$$- f_{13}(x) = x^4 + x + 1$$

$$\begin{array}{r|l} x^4+x^3+x^2+x+1 & x+1 \\ \hline x^4+x^3 & \\ \hline x^2+x+1 & \\ -x^2-x & \\ \hline 1 & \end{array}$$

$$\begin{array}{r|l} x^4+x^3+x^2+x+1 & x^2+1 \\ \hline x^4+x^3 & \\ \hline x^2+x+1 & \\ -x^2-x & \\ \hline 1 & \end{array}$$

$$\begin{array}{r|l} x^4+x^3+x^2+x+1 & x^2+x+1 \\ \hline x^4+x^3+x^2 & \\ \hline x+1 & \end{array}$$

$f_{16}(x) = x^4+x^3+x^2+x+1 \rightarrow$ keltirilmaydigan kophad.

Misol 2.

$$F_2[x] / (x^3+x^2+x)$$

$$p^n = 2^3 = 8$$

$$[0], [1], [x], [x+1], [x^2], [x^2+1], [x^2+x], [x^2+x+1]$$

"+"	0	1	x	x+1	x ²	x ² +1	x ² +x	x ² +x+1
0	0	1	x	x+1	x ²	x ² +1	x ² +x	x ² +x+1
1	1	0	x+1	x	x ² +1	x ²	x ² +x+1	x ² +x
x	x	x+1	0	1	x ² +x	x ² +x+1	x ²	x ² +1
x+1	x+1	x	1	0	x ² +x+1	x ² +x	x ² +1	x ²
x ²	x ²	x ² +1	x ² +x	x ² +x+1	0	1	x	x+1
x ² +1	x ² +1	x ²	x ² +x+1	x ² +x	1	0	x+1	x
x ² +x	x ² +x	x ² +x+1	x ²	x ² +1	x	x+1	0	1
x ² +x+1	x ² +x+1	x ² +x	x ² +1	x ²	x+1	x	1	0

$f(x)$	0	1	x	$x+1$	x^2	x^2+1	x^2+x	x^2+x+1
0	0	0	0	0	0	0	0	0
1	0	1	x	$x+1$	x^2	x^2+1	x^2+x	x^2+x+1
x	0	x	x^2	x^2+x	x^2+x	x^2	x	0
$x+1$	0	$x+1$	x^2+x	x^2+1	x^2	1	x^2	x^2+x+1
x^2	0	x^2	x^2+x	x^2	x	x^2+x	x^2	0
x^2+1	0	x^2+1	x^2	1	x^2+x	$x+1$	x	x^2+x+1
x^2+x	0	x^2+x	x	x^2	x^2	x	x^2+x	0
x^2+x+1	0	x^2+x+1	0	x^2+x+1	0	x^2+x+1	0	x^2+x+1

$$[x] \cdot [x^2] = x^3$$

$$\begin{array}{r} x^3 \mid x^3+x^2+x \\ \underline{x^3+x^2+x} \\ 1 \end{array}$$

$$\text{mod } 2 \rightarrow x^2+x$$

$$x \cdot (x^2+1) = x^3+x$$

$$\begin{array}{r} x^3+x \mid x^3+x^2+x \\ \underline{x^3+x^2+x} \\ 1 \end{array}$$

$$[x^2]$$

$$x \cdot (x^2+x) = x^3+x^2$$

$$\begin{array}{r} x^3+x^2 \mid x^3+x^2+x \\ \underline{x^3+x^2+x} \\ 1 \end{array}$$

$$[x]$$

$$x(x^2+x+1) = x^3+x^2+x$$

$$(x+1) \cdot (x+1) = x^2+2x+1$$

$$\text{mod } 2 \rightarrow x^2+1$$

$$(x+1) \cdot x^2 = x^3+x$$

$$\begin{array}{r} x^3+x \mid x^3+x^2+x \\ \underline{x^3+x^2+x} \\ 1 \end{array}$$

$$[x^2]$$

$$(x+1)(x^2+1) = x^3+x^2+x+1$$

$$\begin{array}{r} x^3+x^2+x+1 \mid x^3+x^2+x \\ \underline{x^3+x^2+x} \\ 1 \end{array}$$

$$[1]$$

$$(x+1)(x^2+x) = x^3+x^2+x^2+x$$

$$\text{mod } 2 \rightarrow x^3+x$$

$$(x^3+x) \pmod{(x^3+x^2+x)} = x^2$$

$$(x+1)(x^2+x+1) = x^3+x^2+x^2+x+x+1$$

$$\text{mod } 2 \rightarrow x^3+1$$

$$(x^3+1) \pmod{(x^3+x^2+x)} = x^2+x+1$$

$$x^2 \cdot x^2 = x^4$$

$$\begin{array}{r|l} x^4 & x^3+x^2+x \\ -x^4+x^3+x^2 & x+1 \\ \hline x^3+x^2 & \\ -x^3+x^2+x & \\ \hline [x] & \end{array}$$

$$x^2 \cdot (x^2+1) = x^4+x^2$$

$$\begin{array}{r|l} x^4+x^2 & x^3+x^2+x \\ -x^4+x^3+x^2 & x+1 \\ \hline x^3 & \\ -x^3 & \\ \hline x^3+x^2+x & \\ [x^2+x] & \end{array}$$

$$x^2 \cdot (x^2+x) = x^4+x^3$$

$$\begin{array}{r|l} x^4+x^3 & x^3+x^2+x \\ -x^4+x^3+x^2 & x \\ \hline [x^2] & \end{array}$$

$$x^2 \cdot (x^2+x+1) = x^4+x^3+x^2$$

$$x^4+x^3+x^2 = x \cdot (x^3+x^2+x)$$

$$(x^2+1)(x^2+1) = x^4+2x^2+1$$

$$\text{mod } 2 \rightarrow x^4+1$$

$$\begin{array}{r|l} x^4+1 & x^3+x^2+x \\ -x^4+x^3+x^2 & x+1 \\ \hline x^3+x^2+1 & \\ -x^3+x^2+x & \\ \hline [x+1] & \end{array}$$

$$(x^2+1) \cdot (x^2+x) = x^4+x^3+x^2+x$$

$$\begin{array}{r|l} x^4+x^3+x^2+x & x^3+x^2+x \\ -x^4+x^3+x^2 & x \\ \hline [x] & \end{array}$$

$$(x^2+1)(x^2+x+1) =$$

$$= x^4+x^2+x^3+x+x^2+1 =$$

$$= x^4+x^3+x+1$$

$$\begin{array}{r|l} x^4+x^3+x+1 & x^3+x^2+x \\ -x^4+x^3+x^2 & x \\ \hline [x^2+x+1] & \end{array}$$

$$(x^2+x)(x^2+x) = x^4+x^3+x^3+x^2$$

$$\text{mod } 2 \rightarrow x^4+x^2$$

$$\begin{array}{r|l} x^4+x^2 & x^3+x^2+x \\ -x^4+x^3+x^2 & x+1 \\ \hline x^3 & \\ -x^3 & \\ \hline x^3+x^2+x & \\ [x^2+x] & \end{array}$$

$$(x^2+x)(x^2+x+1) =$$

$$= x^4+x^3+x^3+x^2+x^2+x$$

$$= x^4+x$$

$$\begin{array}{r|l} x^4+x & x^3+x^2+x \\ -x^4+x^3+x^2 & x+1 \\ \hline x^3+x^2+x & \\ -x^3+x^2+x & \\ \hline [x] & \end{array}$$

$$(x^2+x+1)(x^2+x+1) = x^4+x^3+x^2+x^3+x^2+x$$

$$+ x^2+x+1 = x^4+x^2+1$$

$$\begin{array}{r|l} x^4+x^2+1 & x^3+x^2+x \\ -x^4+x^3+x^2 & x+1 \\ \hline x^3+1 & \\ -x^3+x^2+x & \\ \hline [x^2+x+1] & \end{array}$$

$F_q[x]$ - halqada keltirilmaydigan n -darajali ko'phadlar sonini aniqlash.

Myobius funksiyasi.

Ta'rif: Myobius funksiyasi deb, shunday $\mu(n)$, $n \in \mathbb{N}$ -ga

$$\mu(n) = \begin{cases} 1, & \text{agar } n=1 \\ (-1)^k, & \text{agar } n\text{-soni } k\text{-ta tub sonlar} \\ & \text{ko'paytmasidan iborat bo'lsa} \\ 0, & \text{agar } n\text{-soni biror tub son} \\ & \text{kvadratiga bo'linsa.} \end{cases}$$

(3-shart bajarilmasa)

Quyidagicha belgilash kiritaylik:

$\sum_{d|n}$ - bu $n \in \mathbb{N}$ - sonining barcha d -butun bo'luvchilari bo'yicha yig'indi.

Teorema 1. $F_q[x]$ - halqadagi n -darajali normallashtirgan keltirilmaydigan ko'phadlar soni:

$$\boxed{N_q(n) = \frac{1}{n} \sum_{d|n} \mu\left(\frac{n}{d}\right) q^d = \frac{1}{n} \sum_{d|n} \mu(d) q^{n/d}}$$

Misol 1. $F_2[x]$ - xalqadagi 4-darajali normallashtgan keltirilmaydigan ko'phadlar soni topilsin.

Yechish: $q=2$, $n=4$, $d=\{1, 2, 4\}$

$$N_q(4) = \frac{1}{4} \left(\mu\left(\frac{4}{1}\right) \cdot 2^1 + \mu\left(\frac{4}{2}\right) \cdot 2^2 + \mu\left(\frac{4}{4}\right) \cdot 2^4 \right) =$$

$$= \frac{1}{4} \left(\mu(4) \cdot 2 + \mu(2) \cdot 4 + \mu(1) \cdot 2^4 \right) =$$

$$= \frac{1}{4} (0 \cdot 2 + (-1) \cdot 4 + 1 \cdot 2^4) = \frac{1}{4} \cdot (16 - 4) = \textcircled{3}$$

GSM da AS algoritmi qo'llanilgan.

Teorema 2. $F_q[x]$ - xalqadagi n -darajali keltirilmaydigan (normallashtgan) ko'phadlar kopaytmasi quyidagi formula orqali topiladi.

$$I(q; n; x) = \prod_{d|n} (x^{q^{n/d}} - x)^{\mu(n/d)} = \prod_{d|n} (x^{q^{n/d}} - x)^{\mu(d)}$$

O'xirifa: ~~F_2~~ F_2 maydonida 8-darajali keltirilmaydigan ko'phadlar soni nechta ($n=8$ yani $N_2(8) = ?$)

Heckish: (1) $n=8$ $N_2(8)=?$

$$N_2(8) = \frac{1}{8} \left(\mu_{\substack{1 \\ 1}}(1) \cdot (2^8)^{8/1} + \mu_{\substack{-1 \\ 1}}(2) \cdot (2^8)^{8/2} + \mu_{\substack{1 \\ 0}}(4) \cdot (2^8)^{8/4} + \mu_{\substack{1 \\ 0}}(8) \cdot (2^8)^{8/8} \right) =$$

$$= \frac{1}{8} (2^{64} - 2^{32}) = \underline{2^{61} - 2^{29}}$$

Jawab: $2^{61} - 2^{29}$

(2) $I(q; n; x) = \prod_{d|n} (x^{q^{n/d}} - x)^{\mu(d)}$

$q=2, n=4, d=\{1; 2; 4\}$

$$I(2; 4; x) = (x^{2^4} - x)^{\mu(1)} \cdot (x^{2^2} - x)^{\mu(2)} \cdot (x^{2^1} - x)^{\mu(4)} =$$

$$= (x^{16} - x)^1 \cdot (x^4 - x)^{-1} \cdot (x^2 - x)^0 = \frac{x^{16} - x}{x^4 - x} = \frac{x^{15} - 1}{x^3 - 1}$$

$q=2, n=8, d=\{1; 2; 4; 8\}$

$$I(2; 8; x) = (x^{2^8} - x)^{\mu(1)} \cdot (x^{2^4} - x)^{\mu(2)} \cdot (x^{2^2} - x)^{\mu(4)} \cdot (x^{2^1} - x)^{\mu(8)} =$$

$$= (x^{256} - x)^1 \cdot (x^{16} - x)^{-1} \cdot (x^4 - x)^0 \cdot (x^2 - x)^0 =$$

$$= \frac{x^{256} - x}{x^{16} - x} = \frac{x^{255} - 1}{x^{15} - 1}$$

$q=2^8, n=8, d=\{1; 2; 4; 8\}$

$$I(2^8; 8; x) = (x^{256^8} - x)^{\mu(1)} \cdot (x^{256^4} - x)^{\mu(2)} \cdot (x^{256^2} - x)^{\mu(4)} \cdot (x^{256} - x)^{\mu(8)} =$$

$$= \frac{x^{256^8} - x}{x^{256^4} - x}$$

Jawab: $\frac{x^{256^8} - x}{x^{256^4} - x}$